PERFORMANCE PROFILING · NUMBA CUDA BACKENDS

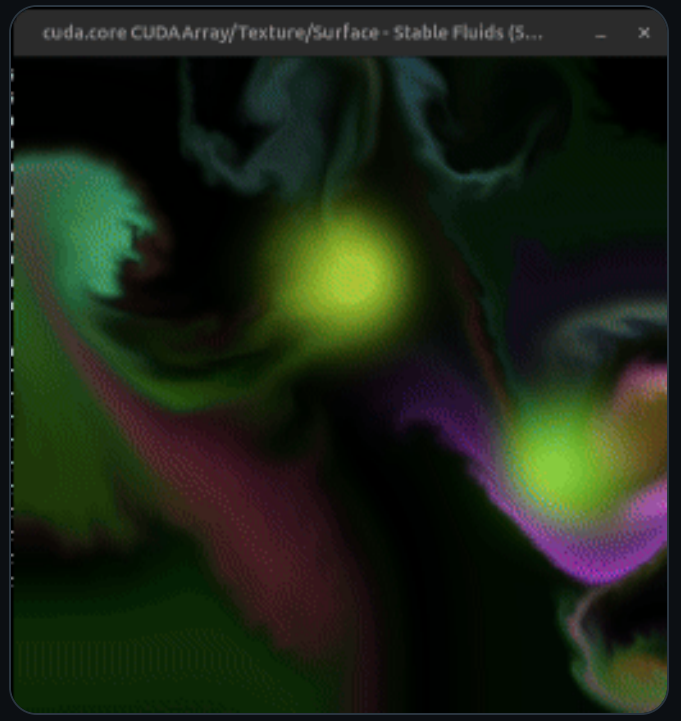
numba-cuda vs numba-cuda-mlir

The question: does the choice of CUDA compiler backend change performance — and where? **The method:** profile the identical Stable-Fluids GPU solver (one source file, 9 kernels) built by both compilers, holding the GPU, CUDA 13.3 toolchain, workload, and Python version constant — so any difference is the compiler alone.

2
COMPILERS · 1 SOURCE

9
KERNELS · 256^2–2048^2

5
MEASUREMENT AXES



the workload — 512^2 Stable-Fluids

Rob Parolin
rparolin@nvidia.com

01 · METHOD — WHAT IS HELD CONSTANT

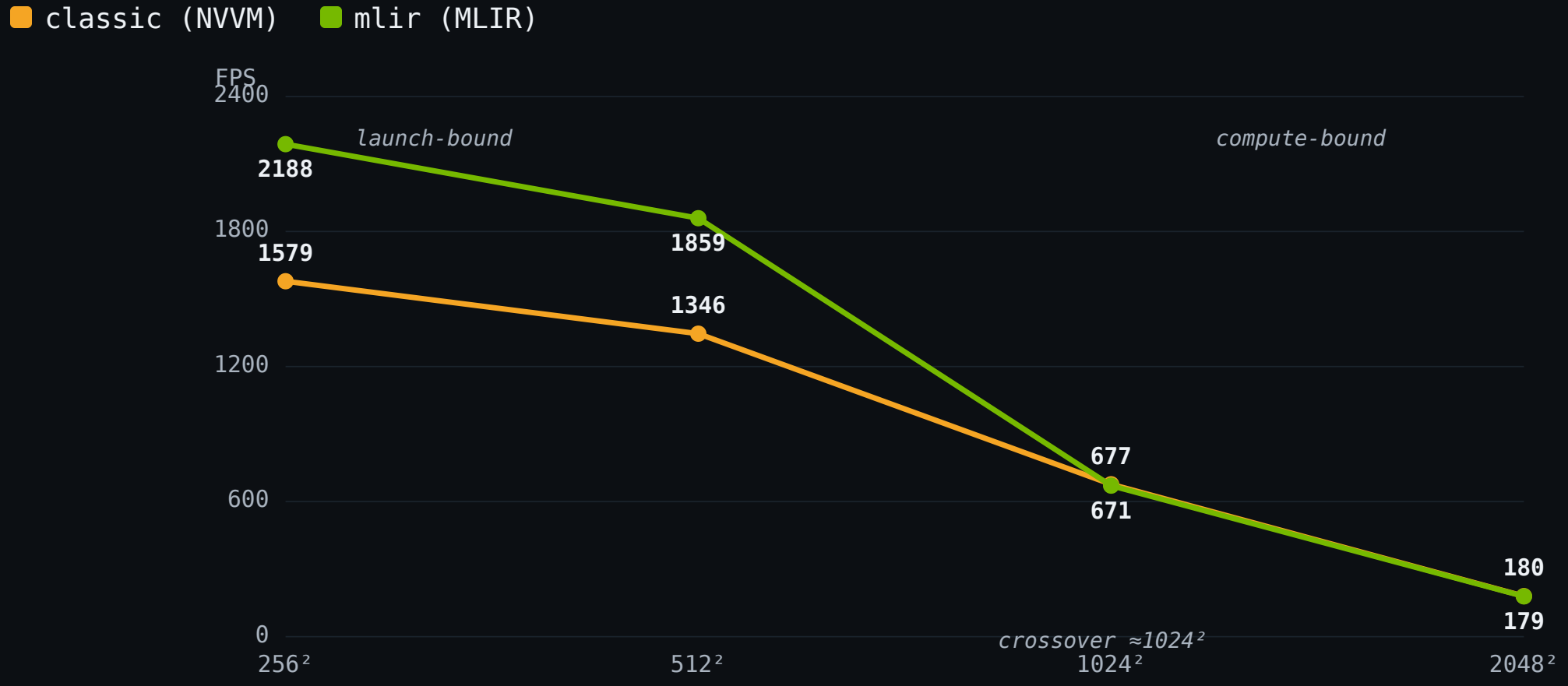
One variable: the compiler frontend

	CLASSIC	MLIR
package	numba-cuda 0.30.2	numba-cuda-mlir 0.4.0
frontend	Numba IR → LLVM → NVVM → PTX	MLIR → PTX (LTOIR)
opt / LTO default	NVVM opt-3, no LTO	LTO, opt>0
Python	3.12.13	3.12.13
cuda-bindings / ptxas	13.3.1 / CUDA 13.3	13.3.1 / CUDA 13.3
GPU	RTX 5880 Ada (sm_89), driver 595.71.05	

Bit-exact output on all 9 kernels at 256^2 / 512^2 / 1024^2 / 2048^2 → a fair race on identical work.

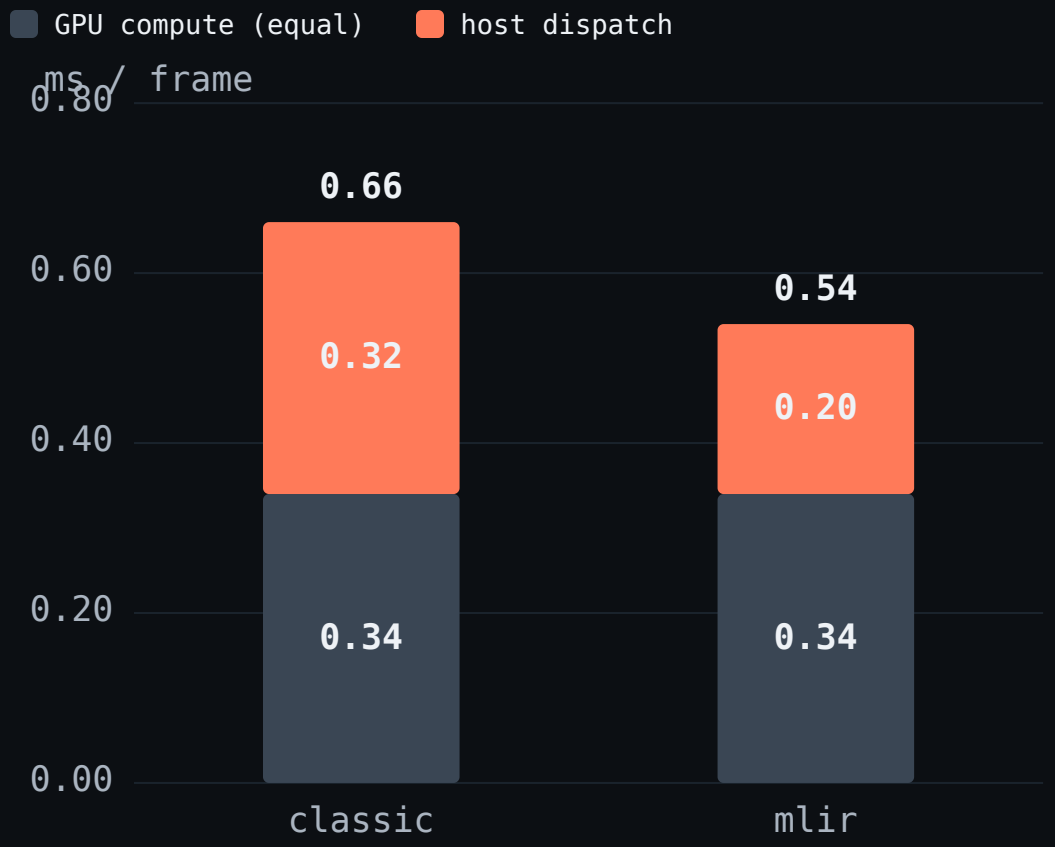
02 · TREND — FPS VS PROBLEM SIZE

The launch-bound → compute-bound crossover



Small grids = **host-launch-bound** (37 launches/frame): mlir wins **~38%**. Big grids = GPU-bound: **identical**.

Equal GPU time; unequal dispatch cost



- The steel base — GPU compute — is the **same 0.34 ms** in both.
- The coral top — host dispatch — differs: **8.6 µs** vs **5.3 µs** per launch.
- × 37 launches/frame builds the total; mlir is **~38% lighter** to dispatch.
- **Why:** classic runs ~2× the Python per launch (**69** vs **34** calls) — re-resolving the CUDA context and re-prepping args every time; mlir caches both.

Where classic spends the extra per-launch time

1. Re-resolve the CUDA context — every launch

```
# cudadrv/devices.py
with self._lock:
  with driver.get_active_context() as ac:   # driver query, each call
    ...
```

2. Re-prep every argument

```
# dispatcher.py _prepare_args
for ext in reversed(self.extensions):      # hook loop
    ty, val = ext.prepare_args(...)
devary = wrap_arg(val).to_device(retr, stream)   # re-wrap + convert
```

3. Safety-wrapped driver calls (2×) + cufunc lookup

```
# cudadrv/driver.py — twice per launch
def safe_cuda_api_call(*a):
    return self._check_cuda_python_error(fname, libfn(*a))
cufunc = self._codelibrary.get_cufunc()   # lookup each launch
```

mlir — cached fast path

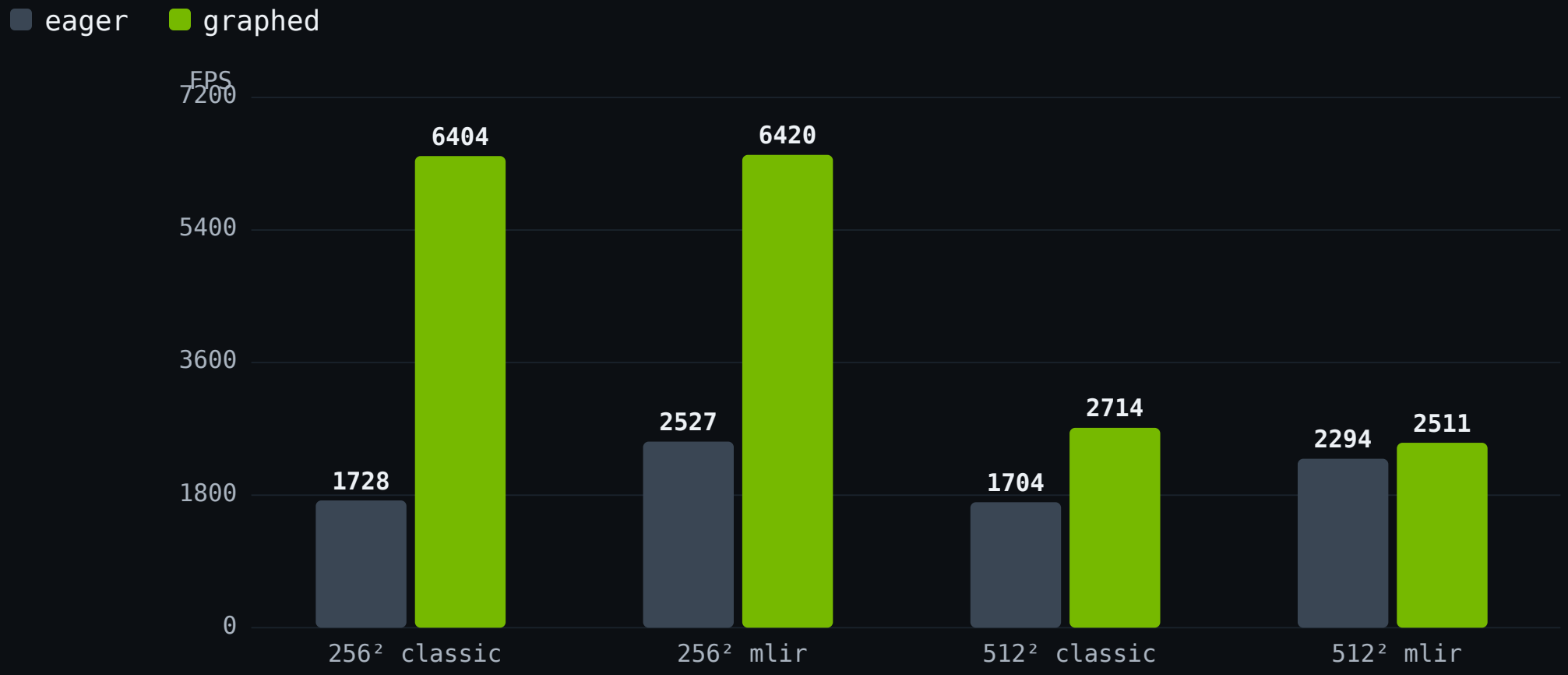
```
# numba_cuda_mlir/descriptor.py
type_key = tuple(type(a) for a in args)
cached = self._sig_cache.get(type_key)
if cached and fast_ok:
    return self._launch(argtypes, args)  # skip re-prep
```

69 vs 34

PYTHON CALLS / LAUNCH · CLASSIC VS MLIR

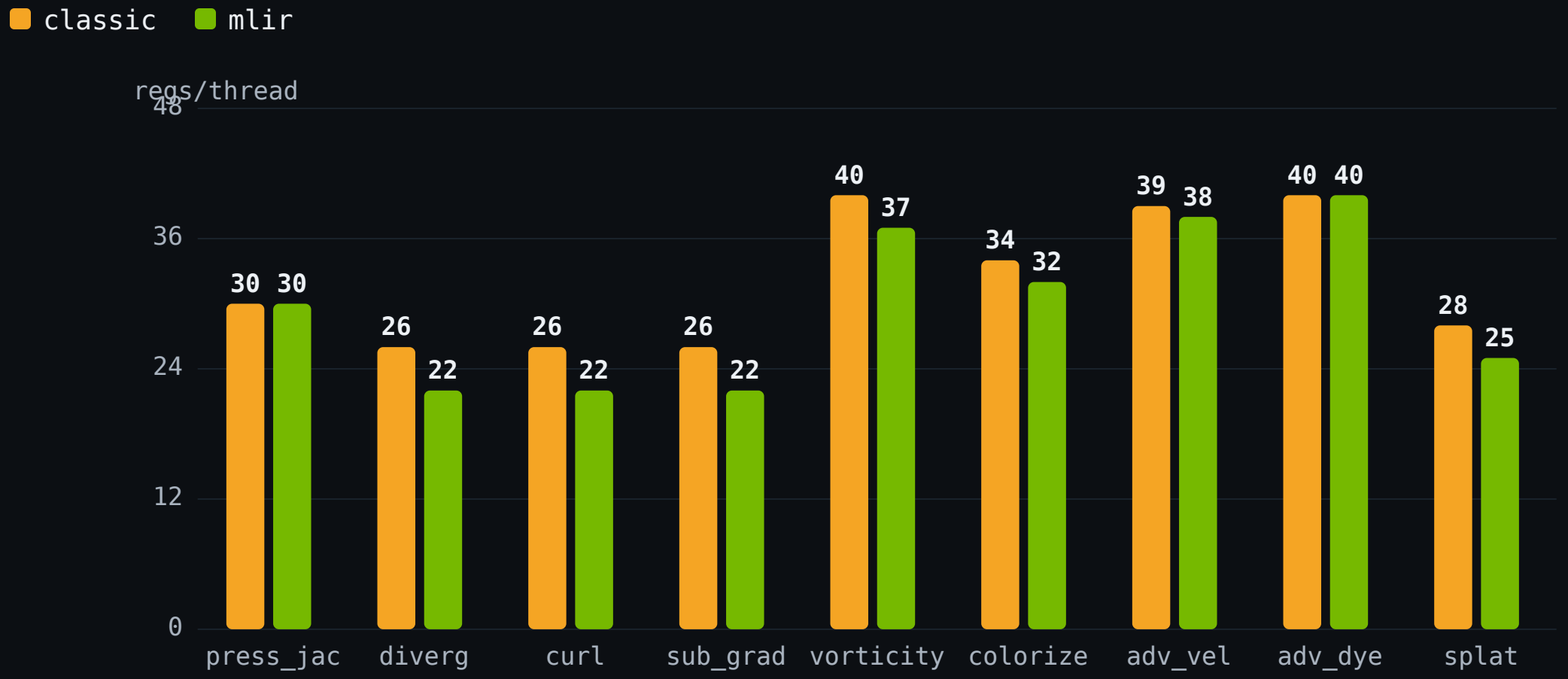
The deeper cause: numba-cuda passes each array by its full kernel calling convention (meminfo · parent · size · itemsize · ptr · shape · strides) — extra host marshaling *and* argument registers per launch; numba-cuda-mlir streamlines this ABI. — mechanism per G. Markall

CUDA graphs erase the gap



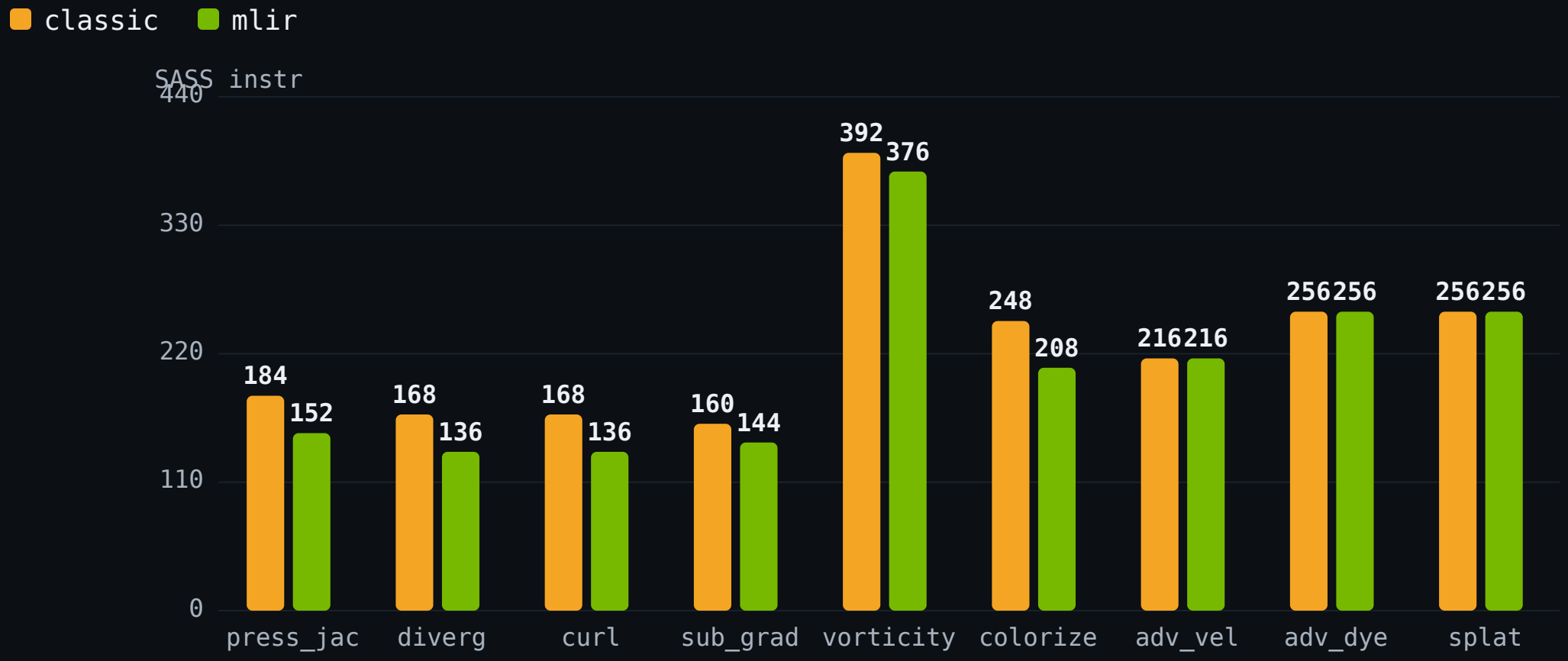
Capture once, replay as a single launch. Both **converge to the same FPS** — classic gains more (**3.7×** vs 2.5×). mlir's eager edge **disappears**.

Slightly fewer registers — and it doesn't matter



All below the 42-reg occupancy limit → **no** occupancy change. Dominant **pressure_jacobi**: 30 = 30.

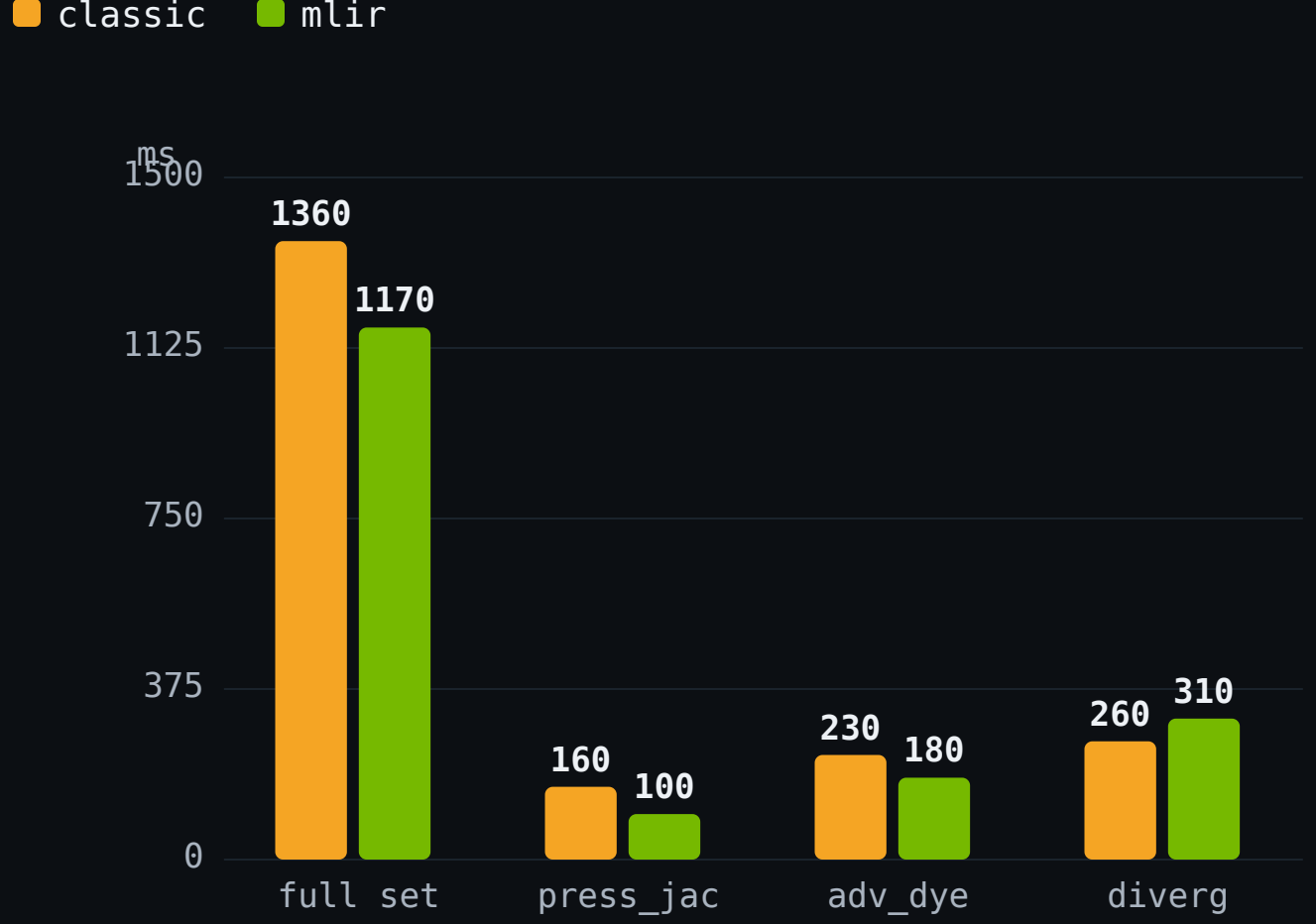
Fewer instructions ≠ faster here



mlir trims −10–19% on stencils, classic wins colorize — yet **GPU time is identical**. The limiter is **float64 throughput** (Python literals promote), not instruction count.

08 · JIT COMPILATION TIME

mlir compiles ~14% faster



1.36s
CLASSIC — FULL SET

1.17s
MLIR — FULL SET (3.12)

Context pre-warmed → pure compile. No on-disk cache, so paid every process.

Match the settings, and the compilers tie

REGISTERS / THREAD pressure_jacobi · fewer = better	LTO OFF	LTO ON
classic	30	30
mlir	50	30

- **LTO** (link-time optimization) is a knob **both** compilers have.
- mlir turns it **on** by default; classic leaves it **off** — that's the whole reason mlir's code looked tighter on slides 6–7.
- Turn LTO on for both → **identical, 30 regs**. Neither compiler is fundamentally better.

A tie on everything — except how fast they launch kernels

WHAT WE MEASURED	WINNER	BY HOW MUCH
output correctness	**tie**	exact same numbers, every grid size
GPU kernel speed	**tie**	same runtime & occupancy
GPU code size	**tie***	mlir writes ~15% fewer instrs — but no faster; equal once both use LTO
compile time	**mlir**	~14% faster
kernel-launch overhead	**mlir**	5.3 vs 8.6 µs/launch → +38% FPS on small grids
FPS on big grids / with graphs	**tie**	identical

> **Core idea:** for performance the two compilers are interchangeable — same results, same kernel speed. mlir's **only** real edge is cheaper kernel launching, which matters on small workloads and **vanishes with CUDA graphs**.

Pick on launch model, not the compiler

- **Same results, same kernel speed** — identical output; equal GPU time and occupancy.
- **mlir launches kernels ~38% cheaper** — 5.3 vs 8.6 µs → +38% FPS below 1024^2.
- **CUDA graphs ≈3× FPS** on both backends — and erase the mlir-vs-classic gap.
- **mlir compiles ~14% faster** — 1.17 s vs 1.36 s for all 9 kernels.
- **Biggest latent win:** kernels run in float64 (Python literals); float32 would cut math on both.